MICROKERNEL ARCHITECTURE MANUAL

HYDRA OS

UNIX-like Distributed Microkernel for Modern UEFI Hardware

BOOT SEQUENCE PIPELINE

UEFI POST → GDT/TSS → CPU Diagnostics → Frame Allocator → PML4 Paging → Capability Space → IPC Rendezvous

Edition: v0.1.0 Architecture Specification

Target Architecture: x86_64-unknown-uefi (Native 64-bit Long Mode)

Security Model: Capability-Based Access Control (seL4-style)

IPC Subsystem: Synchronous & Asynchronous Message Rendezvous

Generated: October 22, 2026

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SYSTEM OVERVIEW • BOOT SEQUENCE ORDER #01

Hydra OS - Codebase Documentation Index

Welcome to the technical documentation for **Hydra OS**. Every source code file in the repository is documented with real-world analogies, conceptual explanations, architectural diagrams, and API breakdowns designed to be accessible to anyone with general programming knowledge.

🗺️ Documentation Directory Map

```
docs/
├── README.md                    # This Index
├── build.md                     # Build automation script (build.rs)
└── src/
    ├── main.md                  # Kernel entry point & bootstrap orchestrator (src/main.rs)
    ├── console.md               # Thread-safe console & formatting macros (src/console.rs)
    ├── arch/
    │   ├── mod.md               # Architecture-specific module root (src/arch/mod.rs)
    │   ├── gdt.md               # Global Descriptor Table, TSS & Rings (src/arch/gdt.rs)
    │   ├── cpu.md               # Control registers & CPU diagnostics (src/arch/cpu.rs)
    │   └── x86_64/
    │       └── boot.md          # Legacy BIOS boot assembly (header.asm, main.asm, main64.asm)
    ├── memory/
    │   ├── mod.md               # Memory management module root (src/memory/mod.rs)
    │   ├── frame_allocator.md   # Physical memory manager & 4 KiB frames (src/memory/frame_all
    │   └── paging.md            # Virtual memory & 4-level PML4 translation (src/memory/paging
    ├── capability/
    │   ├── mod.md               # Capability types & Rights bitflags (src/capability/mod.rs)
    │   └── table.md             # Task capability table & CSpace (src/capability/table.rs)
    ├── shell.md                 # Interactive microkernel shell & command REPL (src/shell.rs)
    └── ipc/
        ├── mod.md               # IPC subsystem root (src/ipc/mod.rs)
        ├── message.md           # Fast register payload & transferred caps (src/ipc/message.rs
        └── endpoint.md          # Message rendezvous channels (src/ipc/endpoint.rs)
```

📚 Module Reference Table

Subsystem	Source File	Documentation	Key Concepts & Analogies
Boot & Main	`src/main.rs`	docs/src/main.md	Entry point, boot sequence, hardware diagnostics, and panic handler.
Console I/O	`src/console.rs`	docs/src/console.md	Thread-safe `print!` / `println!` macros, keyboard input, screen formatting.
Interactive Shell	`src/shell.rs`	docs/src/shell.md	Command REPL, real-time memory/CPU inspection, CSpace manipulation, IPC testing.
GDT & TSS	`src/arch/gdt.rs`	docs/src/arch/gdt.md	Ring 0 / Ring 3 privilege levels, Task State Segment, double-fault stack.
CPU Control	`src/arch/cpu.rs`	docs/src/arch/cpu.md	Control registers (`CR0`, `CR2`, `CR3`, `CR4`), paging roots.
Arch Root	`src/arch/mod.rs`	docs/src/arch/mod.md	Architecture abstraction layer for hardware independence.
Physical Memory	`src/memory/frame_allocator.rs`	docs/src/memory/frame_allocator.md	UEFI memory map parsing, 4 KiB frame allocator ("parking lot attendant").
Virtual Memory	`src/memory/paging.rs`	docs/src/memory/paging.md	4-level x86_64 paging tree

Subsystem	Source File	Documentation	Key Concepts & Analogies
			(PML4), address translation, page protection.
Memory Root	`src/memory/mod.rs`	docs/src/memory/mod.md	Physical and virtual memory coordinator.
Capability Core	`src/capability/mod.rs`	docs/src/capability/mod.md	Security keycards (`Rights::READ` , `WRITE` , `EXECUTE` , `GRANT` , `CALL`).
Capability Table	`src/capability/table.rs`	docs/src/capability/table.md	Per-task CSpace keyring, permission checking, access revocation.
IPC Message	`src/ipc/message.rs`	docs/src/ipc/message.md	Registered courier package: 4 fast inline register arguments + security token.
IPC Endpoint	`src/ipc/endpoint.rs`	docs/src/ipc/endpoint.md	Postal mailbox rendezvous point, FIFO circular ring buffer.
IPC Root	`src/ipc/mod.rs`	docs/src/ipc/mod.md	Inter-process communication module export.
Build Automation	`build.rs`	docs/build.md	NASM assembly build script and link configuration.
Legacy Boot	`src/arch/x86_64/boot/`	docs/src/arch/x86_64/boot.md	Multiboot2 header and 32-bit to 64-bit assembly trampoline.

🚀 Boot Sequence Chronological Topic Order

When understanding or reviewing Hydra OS from initial hardware power-on through microkernel readiness, the documentation is structured sequentially:

1. **Firmware Handoff & Bootstrapping**: - Legacy Boot Trampoline & Long Mode Switch (Multiboot2 header, 32-bit to 64-bit Long Mode switch) - Build Automation & Assembly Pipeline (NASM toolchain and UEFI link settings)
2. **UEFI Entry & Bootstrap Orchestration**: - Kernel Entry Point & Bootstrap Pipeline (UEFI `main()`, panic handler, subsystem coordination, shell handover)
3. **Early Diagnostic Output & Keyboard Input**: - UEFI Console & Formatting (`print!`, `println!`, thread-safe `CONSOLE` spinlock, keyboard polling, cursor management)
4. **CPU Architecture & Hardware Isolation (Step 1 & 2)**: - Architecture Abstraction Layer (Hardware abstraction root) - Global Descriptor Table (GDT) & TSS (Ring 0 / Ring 3 privilege levels, Task State Segment, Double Fault IST stack) - CPU Control Registers & Diagnostics (CR0 protection, CR2 fault addresses, CR3 PML4 paging root, CR4 extensions)
5. **Physical & Virtual Memory Management (Step 3 & 4)**: - Memory Subsystem Coordinator (Memory management root) - Physical Memory & Frame Allocator (UEFI memory map parsing, 1 MiB low memory reservation, 4 KiB frame allocator) - Virtual Memory & 4-Level Paging (x86_64 PML4 translation, 2 MiB / 1 GiB huge page decoding)
6. **Capability Security Model (Step 5)**: - Capability Security Primitives (Security keycards, `Rights` bitflags, `CapType` kernel resource identifiers) - Task Capability Space & Table (Per-task CSpace table, fixed 256-slot array storage, rights verification)
7. **Inter-Process Communication (Step 6)**: - IPC Subsystem Root (Microkernel message passing architecture) - IPC Fast Register Messages (4 fast inline register payload words, labels, transferred capability tokens) - IPC Endpoints & Mailbox Rendezvous (Circular FIFO ring buffers, rendezvous channels, thread-safe `SharedEndpoint`)
8. **Interactive User Space & Subsystem Diagnostics**: - Interactive Microkernel Shell (Command REPL, real-time memory/CPU inspection, CSpace manipulation, IPC testing, RTC clock, reboot/shutdown)

📄 Generating the Unified Documentation PDF

You can generate a single, professional PDF manual containing all documentation files, automatically formatted and ordered by the boot sequence topics:

Windows:

```
generate_pdf.bat
```

or

```
python scripts\generate_docs_pdf.py
```

Linux / WSL / macOS:

```
make docs-pdf
```

or

```
python3 scripts/generate_docs_pdf.py
```

The compiled PDF is generated at `docs/Hydra_OS_Documentation.pdf` and mirrored to `dist/Hydra_OS_Documentation.pdf`. An HTML version suitable for browser inspection or custom printing is also created at `docs/Hydra_OS_Documentation.html`.

BUILD & TOOLCHAIN • BOOT SEQUENCE ORDER #02

Build Script Documentation

- **Source File**: `build.rs`
- **Subsystem**: Toolchain & Assembly Build Automation

---

1. Purpose & Context

In earlier phases or legacy configurations, Hydra supported a hybrid BIOS/Multiboot bootloader mode that required compiling 16-bit and 32-bit assembly files (`header.asm` , `main.asm` , `main64.asm`) with the `nasm` assembler before linking into ELF objects.

In current native UEFI builds, Cargo compiles Rust directly targeting `x86_64-unknown-uefi` without requiring external assembly compilation, with `build = false` in `Cargo.toml` .

---

2. Key Functions

`assemble_file(file: &str, output_dir: &str)`

1. Checks the `DEBUG` environment variable to determine whether to emit DWARF debug symbols (`-g`).
2. Invokes `nasm` to compile an `.asm` source file into an ELF64 object (`.o`).
3. Instructs Cargo to re-run the build script if the source assembly file changes (`cargo:rerun-if-changed`).
4. Tells the Rust linker to link the resulting object file (`cargo:rustc-link-object`).

`assemble_all(output_dir: &str)`

Loops through the standard assembly boot files: - `boot/header.asm` : Multiboot2 header definition. - `boot/main.asm` : 32-bit protected mode entry point, page table setup, and long mode switch. - `boot/main64.asm` : 64-bit trampoline to Rust entry point.

STAGE 0: FIRMWARE HANDOFF • BOOT SEQUENCE ORDER #03

Legacy Assembly Bootstrapper Documentation

- **Source Directory**: `src/arch/x86_64/boot/`
- **Files**:
- `header.asm`
- `main.asm`
- `main64.asm`
- **Subsystem**: Architecture / Legacy BIOS Boot Trampoline

---

1. Overview

Before migrating to native **UEFI**, x86 PCs booted via legacy **BIOS**. Under BIOS: 1. The CPU boots in **16-bit Real Mode**. 2. A bootloader (like GRUB) loads the kernel into **32-bit Protected Mode**. 3. The kernel itself is responsible for setting up initial page tables, enabling PAE, switching to **64-bit Long Mode**, and jumping to 64-bit code.

These assembly files implement that transition for reference or hybrid BIOS/GRUB boot media.

---

2. File Breakdown

1. `header.asm` (Multiboot2 Header)

- Defines the magic numbers and architecture tags required by Multiboot2-compliant bootloaders (such as GRUB).
- Allows GRUB to recognize the kernel file as bootable.

2. `main.asm` (32-Bit to 64-Bit Transition)

- **Stack Initialization**: Allocates an initial boot stack.
- **CPU Checks**:
- `check_multiboot`: Verifies that the bootloader passed the expected Multiboot magic value in `EAX`.
- `check_cpuid`: Verifies that the CPU supports the `cpuid` instruction.
- `check_long_mode`: Queries CPUID to verify 64-bit Long Mode hardware support.
- **Identity Paging**: Configures an initial 2 MiB identity-mapped page table so memory addresses map 1:1 during the switch.
- **Enabling Long Mode**: Sets the Physical Address Extension bit (`PAE`) in `CR4`, enables Long Mode in the Extended Feature Enable Register (`EFER` MSR), and enables Paging in `CR0`.
- **GDT Setup**: Loads a temporary 64-bit code segment and performs a far jump to `long_mode_start`.

3. `main64.asm` (64-Bit Trampoline)

- Re-zeros data segment registers (`DS` , `ES` , `FS` , `GS` , `SS`).
- Invokes the Rust 64-bit entry point (`call kernel_main`).
- Halts the CPU if the function ever returns.

STAGE 1: KERNEL ENTRY POINT • BOOT SEQUENCE ORDER #04

Kernel Entry Point & Bootstrapping Documentation

- **Source File**: `src/main.rs`
- **Subsystem**: Microkernel Bootstrap & Initialization Orchestrator

---

1. Conceptual Overview

`src/main.rs` is the primary entry point executed when the UEFI firmware finishes hardware Power-On Self-Test (POST) and transfers control to `BOOTX64.EFI`.

Because Hydra targets native 64-bit UEFI directly: - The CPU is already in **64-bit Long Mode**. - Paging is enabled by the UEFI firmware. - The kernel receives a valid `uefi::prelude::SystemTable` reference, enabling it to access firmware services before taking full ownership of the machine.

---

2. Boot Execution Pipeline

```
graph TD
    A[UEFI Firmware POST] ⟶ B[main entry point]
    B ⟶ C[Initialize Console & Clear Screen]
    C ⟶ D[arch::gdt::init: Load GDT, TSS, Ring 0/3 Segments]
    D ⟶ E[arch::cpu::read_state: Inspect CR0, CR3 PML4 Root, CR4]
    E ⟶ F[memory::frame_allocator: Parse UEFI Memory Map & Test Frame]
    F ⟶ G[memory::paging: Test PML4 Virtual-to-Physical Translation]
    G ⟶ H[capability: Setup Task CSpace & Capability Table]
    H ⟶ I[ipc: Test Endpoint Message Rendezvous]
    I ⟶ J[Halt Loop with Interrupts Disabled]
```

---

3. Step-by-Step Code Walkthrough

The Panic Handler

```
#[panic_handler]
fn panic(_info: &PanicInfo) -> ! {
    console::set_color(Color::LightRed, Color::Black);
    println!("\n[!] KERNEL PANIC: {:?}", _info);
    x86_64::instructions::interrupts::disable();
    loop {
        x86_64::instructions::hlt();
    }
}
```

If an assertion fails or an unexpected unrecoverable condition occurs, the panic handler prints diagnostic error messages in red to the UEFI console, disables interrupts to prevent re-entrancy, and halts the CPU in a low-power

infinite loop (`hlt`).

Step 1: Segmentation & Hardware Privilege Setup

```
console::set_color(Color::Yellow, Color::Black);
println!("[*] Step 1: Initializing GDT, TSS & Ring 3 Segments...");
arch::gdt::init();
console::set_color(Color::LightGreen, Color::Black);
println!("[+] GDT loaded successfully.");
```

Disables interrupts and installs Hydra's custom Global Descriptor Table and Task State Segment, configuring the CPU for both supervisor (Ring 0) and isolated userland (Ring 3) execution.

Step 2: CPU Diagnostics

```
let cpu = arch::cpu::read_state();
println!("    - CR0: {:?}", cpu.cr0);
println!("    - CR3 (Active PML4 Root): {:#x}", cpu.cr3_frame.as_u64());
println!("    - CR4: {:?}", cpu.cr4);
```

Reads control registers to log the active hardware configuration, including the physical address of the initial root page table (`CR3`).

Step 3: Physical Memory Discovery & Frame Allocation

```
console::set_color(Color::Yellow, Color::Black);
println!("\n[*] Step 2: Discovering Memory & Initializing Frame Allocator...");
memory::frame_allocator::init_memory_from_uefi();

let (usable_mb, allocated_kb) = {
    let alloc = memory::frame_allocator::FRAME_ALLOCATOR.lock();
    (
        alloc.total_usable_memory() / (1024 * 1024),
        alloc.allocated_memory() / 1024,
    )
};
```

Queries UEFI firmware for the memory map, identifies all conventional RAM regions (skipping the first 1 MiB to protect BIOS/IVT data and null pointers), and initializes the kernel's physical frame allocator. It then tests allocating a sample 4 KiB physical frame.

Step 4: Virtual Memory Verification

```
console::set_color(Color::Yellow, Color::Black);
println!("\n[*] Step 3: Verifying 4-Level Paging (PML4) Address Translation...");
let address_space = memory::paging::AddressSpace::active();
let sample_virt = VirtAddr::new(main as *const () as usize as u64);
let sample_phys = unsafe { address_space.translate_addr(sample_virt) };
```

Walks the active 4-level PML4 page table hierarchy to verify that virtual addresses successfully resolve to physical memory frames, accommodating 4 KiB, 2 MiB, and 1 GiB page mappings.

Step 5: Capability Table Initialization

```
console::set_color(Color::Yellow, Color::Black);
println!("\n[*] Step 4: Initializing Task Capability Space...");
let mut cap_table = CapabilityTable::new();
let root_cap = Capability::new(
    CapType::AddressSpace {
        root_cr3: cpu.cr3_frame.as_u64(),
    },
    Rights::ALL,
);
let slot0 = cap_table.insert(root_cap).unwrap();
let ep_cap = Capability::new(
    CapType::Endpoint { endpoint_id: 1 },
    Rights(Rights::READ.0 | Rights::WRITE.0 | Rights::CALL.0),
);
let slot1 = cap_table.insert(ep_cap).unwrap();
```

Establishes the process capability table (CSpace), assigning granular access tokens to kernel resources (Root CR3 address space and IPC Endpoint #1).

Step 6: Microkernel IPC Rendezvous

```
console::set_color(Color::Yellow, Color::Black);
println!("\n[*] Step 5: Testing Microkernel IPC Rendezvous...");
let mut endpoint = Endpoint::new(1);
let msg = Message::with_args(0x1001, [42, 1337, 0, 0]).with_cap(ep_cap);
endpoint.send(msg).unwrap();
if let Some(received) = endpoint.receive() { ... }
```

Demonstrates inter-process communication: sending a message with inline arguments and an attached delegated capability, followed by receiving and consuming the message from the endpoint queue.

Step 7: Subsystems Operational & Terminal Halt Loop

```
console::set_color(Color::LightCyan, Color::Black);
println!("\n═══════════════════════════════════════════════════════════");
println!(" [+] HYDRA MICROKERNEL PHASE 1 SUBSYSTEMS OPERATIONAL!      ");
println!("═══════════════════════════════════════════════════════════");

x86_64::instructions::interrupts::disable();
loop {
    x86_64::instructions::hlt();
}
```

Upon successful boot and subsystem diagnostics, the kernel enters a halted state awaiting hardware interrupts or process scheduling.

STAGE 2: EARLY DIAGNOSTICS • BOOT SEQUENCE ORDER #05

Console & Diagnostic Output Documentation

- **Source File**: `src/console.rs`
- **Subsystem**: Core I/O & Formatting

---

1. Conceptual Overview

In a bare-metal `#![no_std]` environment, Rust's standard library (`std::io`, `std::println!`) does not exist because there is no underlying operating system yet.

The `console` module bridges Rust's `core::fmt` formatting system directly to UEFI's standard text output protocol (`SimpleTextOutput`), providing thread-safe `print!` and `println!` macros that can be used everywhere across the kernel.

---

2. Architecture & Concurrency

```
Any Kernel Thread / Subsystem
            |
            v
println!("Text: {}", value)
            |
            v
     CONSOLE (Mutex)  <--- Thread-safe synchronization via spin::Mutex
            |
            v
   uefi::system::with_stdout
            |
            v
   Physical Video Display / UEFI Console Screen
```

- `spin::Mutex`: Used instead of standard OS mutexes because interrupts or threads can acquire the lock without requiring OS scheduling services.

---

3. API Reference & Methods

`print!` and `println!` Macros

Drop-in replacements for standard Rust printing macros.

```
print!("Initializing subsystem ... ");
println!("Done: {} frames free.", free_count);
```

`console::set_color(foreground: Color, background: Color)`

Changes the active text color and background color on the UEFI screen.

```
use uefi::proto::console::text::Color;

console::set_color(Color::LightGreen, Color::Black);
println!("[+] Operation Succeeded");
```

`console::clear_screen()`

Clears the active screen buffer and resets the cursor to the top-left coordinate `(0, 0)`.

`console::CONSOLE` and `_print`

```
pub static CONSOLE: Mutex<Console> = Mutex::new(Console);

#[doc(hidden)]
pub fn _print(args: fmt::Arguments) {
    let mut console = CONSOLE.lock();
    let _ = console.write_fmt(args);
}
```

`CONSOLE` is a globally accessible, mutex-locked unit struct implementing `core::fmt::Write`. The `print!` and `println!` macros format arguments and invoke `_print`, acquiring the lock to safely serialize output to the UEFI `SimpleTextOutput` protocol.

STAGE 3: CPU ARCHITECTURE • BOOT SEQUENCE ORDER #06

Architecture Module Overview

- **Source File**: `src/arch/mod.rs`
- **Subsystem**: Architecture-Specific Primitives (x86_64)

---

1. Role in Hydra OS

The `arch` module isolates all processor-specific and machine-dependent code away from the rest of the generic microkernel.

When Hydra is ported to other CPU architectures (such as ARM64 or RISC-V), only the implementations inside `arch` need to be swapped out.

---

2. Submodules

- `gdt.rs`: Manages the Global Descriptor Table, Ring 0 / Ring 3 segmentation, and Task State Segment (TSS).
- `cpu.rs`: Provides interfaces for reading x86 control registers (`CR0` , `CR2` , `CR3` , `CR4`).

STAGE 3: CPU ARCHITECTURE • BOOT SEQUENCE ORDER #07

Global Descriptor Table (GDT) & TSS Documentation

- **Source File**: `src/arch/gdt.rs`
- **Subsystem**: Architecture / CPU Privilege & Segmentation

---

1. Conceptual Overview

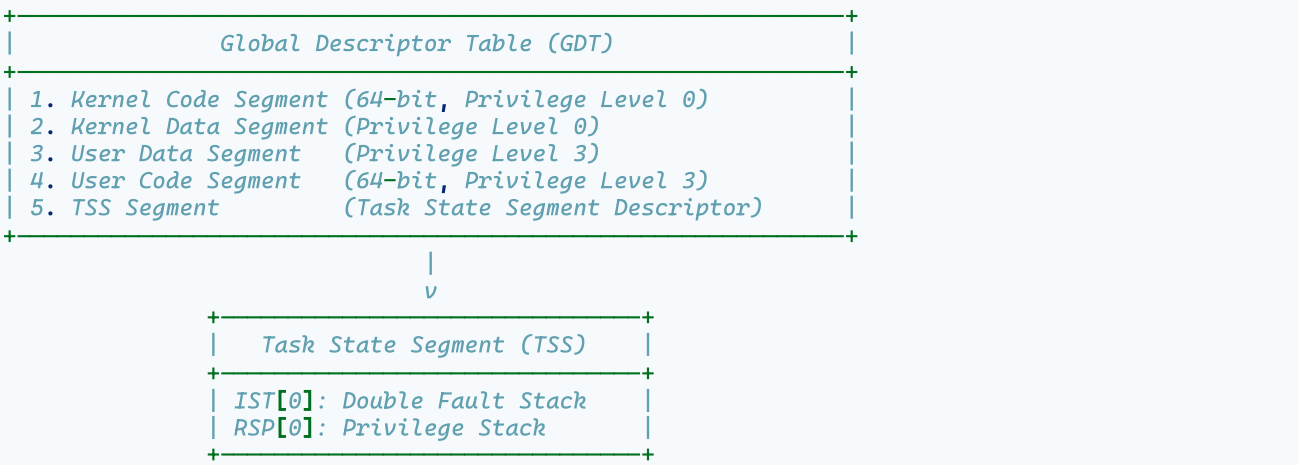
On x86_64 processors, the **Global Descriptor Table (GDT)** is a legacy hardware structure that remains vital for two critical operating system responsibilities: 1. **Privilege Levels (Rings)**: Defining segments for **Ring 0 (Kernel Mode)** and **Ring 3 (Userland Mode)**. 2. **Task State Segment (TSS)**: Defining emergency hardware stack pointers for CPU privilege transitions and catastrophic fault recovery (Double Faults).

Privilege Rings

- **Ring 0**: Full CPU privilege. The kernel executes here and can run privileged instructions (`cli` , `sti` , `invlpg` , write to control registers `CR0..CR4`).
- **Ring 3**: Sandboxed user mode. Applications run here and are restricted by the MMU and capability system.

---

2. Architecture & Components



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3. Data Structures & Stacks

Stack Allocation

```
const STACK_SIZE: usize = 4096 * 5; // 20 KiB
static mut DOUBLE_FAULT_STACK: [u8; STACK_SIZE] = [0; STACK_SIZE];
static mut PRIVILEGE_STACK: [u8; STACK_SIZE] = [0; STACK_SIZE];
```

- `DOUBLE_FAULT_STACK`: When a double fault occurs (e.g. a page fault inside a kernel stack overflow), the CPU cannot use the current corrupted stack. Using the **Interrupt Stack Table (IST index 0)**, the CPU automatically switches to this clean stack.
- `PRIVILEGE_STACK`: When a Ring 3 user program triggers a system call or interrupt to enter Ring 0, the CPU loads this stack into `RSP` (`tss.privilege_stack_table[0]`).

`Selectors`

Stores the segment selectors returned when the segments are appended to the GDT:

```
pub struct Selectors {
    pub kernel_code_selector: SegmentSelector,
    pub kernel_data_selector: SegmentSelector,
    pub user_data_selector: SegmentSelector,
    pub user_code_selector: SegmentSelector,
    pub tss_selector: SegmentSelector,
}
```

---

4. API Reference & Methods

`arch::gdt::init()`

Initializes and prepares Hydra's segmentation and hardware privilege structures in memory: 1. **Prepares Task State Segment (TSS)**: Configures `interrupt_stack_table[0]` for the double-fault emergency stack and `privilege_stack_table[0]` for Ring 3 → Ring 0 privilege transitions. 2. **Builds the GDT**: Allocates and configures Kernel Code, Kernel Data, User Data, User Code, and TSS descriptors. 3. **UEFI Firmware Coexistence**: While native UEFI Boot Services are active, the firmware's IDT relies on firmware code selectors (e.g., `0x38` on VirtualBox EDK2). Hydra ensures GDT and TSS structures are initialized and accessible via `get_selectors()` without disrupting the firmware's active interrupt vectors.

```
arch::gdt::init();
```

`arch::gdt::get_selectors() → &'static Selectors`

Provides access to the active segment selectors (`kernel_code_selector` , `kernel_data_selector` , `user_data_selector` , `user_code_selector` , `tss_selector`) for subsequent userland process spawning, context switches, and interrupt returns.

STAGE 3: CPU ARCHITECTURE • BOOT SEQUENCE ORDER #08

CPU Control & Register Diagnostics Documentation

- **Source File**: `src/arch/cpu.rs`
- **Subsystem**: Architecture / CPU Control Registers

---

1. Conceptual Overview

On x86_64 processors, the processor's operating mode, memory translation mechanisms, and security features are governed by specialized **Control Registers** (`CR0` , `CR2` , `CR3` , `CR4`):

- **CR0**: Controls basic execution features: Protected Mode, Paging enable, Write Protect (prevents kernel from inadvertently overwriting read-only user pages).
- **CR2**: Stores the linear virtual address that triggered the most recent **Page Fault** exception. Vital for demand paging and memory debugging.
- **CR3**: Points to the base physical address of the Level 4 Page Table (PML4). Loading a new value into CR3 immediately switches active virtual address spaces.
- **CR4**: Controls advanced architecture features: Physical Address Extension (PAE), Page Global Enable (PGE), OS support for SSE/AVX floating point, and User-Mode Instruction Prevention (UMIP).

---

2. The `CpuState` Struct

```rust
#[derive(Debug, Clone, Copy)]
pub struct CpuState {
    pub cr0: Cr0Flags,
    pub cr2: u64,
    pub cr3_frame: PhysAddr,
    pub cr4: Cr4Flags,
}
```

Captures an instantaneous snapshot of the machine's hardware configuration registers.

---

3. API Reference

`arch::cpu::read_state() → CpuState`

Queries the CPU hardware registers and packages them into a `CpuState` diagnostic struct.

```rust
use crate::arch::cpu;

let cpu = cpu::read_state();
println!("Paging root physical address: {:#x}", cpu.cr3_frame.as_u64());
```

STAGE 4: MEMORY SUBSYSTEM • BOOT SEQUENCE ORDER #09

Memory Subsystem Overview

- **Source File**: `src/memory/mod.rs`
- **Subsystem**: Memory Management

---

1. Role in Hydra OS

The `memory` module coordinates all physical and virtual memory resources required by the kernel and user processes.

It abstracts low-level hardware memory mechanisms (UEFI memory descriptors, x86_64 paging tables, MMU control registers) into safe, modular Rust interfaces.

---

2. Submodules

- `frame_allocator.rs`: Manages physical memory frames (4 KiB) discovered from the UEFI firmware memory map.
- `paging.rs`: Manages virtual memory address spaces and implements 4-level PML4 address translation.

STAGE 4: MEMORY SUBSYSTEM • BOOT SEQUENCE ORDER #10

Physical Memory & Frame Allocator Documentation

- **Source File**: `src/memory/frame_allocator.rs`
- **Subsystem**: Memory Management (Physical Layer)

---

1. Conceptual Overview

When a computer boots up, its physical Random Access Memory (RAM) is not one continuous empty space. Certain chunks are occupied by motherboard ROM, video graphics RAM, ACPI tables, and UEFI firmware code.

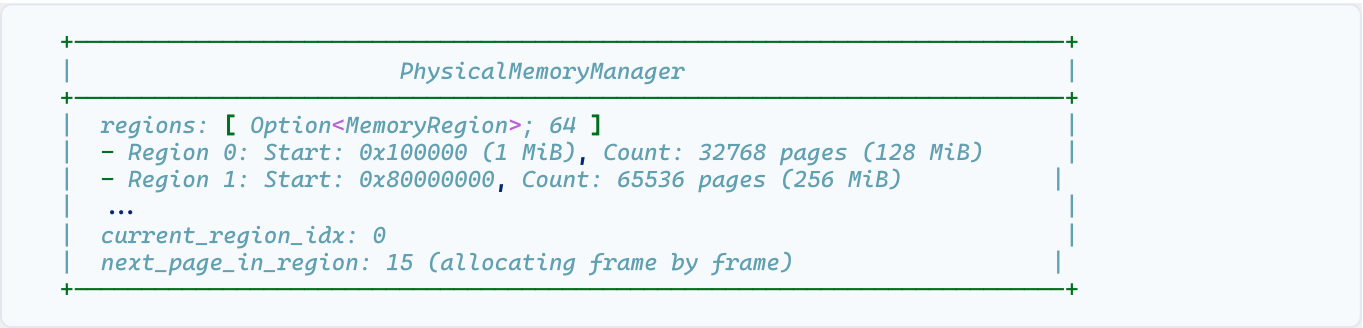
The **Physical Memory Frame Allocator** is responsible for: 1. Asking UEFI firmware which memory regions are safe and usable (`CONVENTIONAL` memory). 2. Dividing that usable RAM into standard **4096-byte (4 KiB) chunks**, called **Physical Frames**. 3. Handing out frames one-by-one to the kernel when it needs memory for page tables, stacks, or user processes.

The Real-World Analogy

Think of RAM as a vast parking lot: - Some spots are reserved for emergency vehicles or building infrastructure (firmware/hardware). - The **Physical Memory Manager** is the parking attendant with a map of all standard open spots. - Whenever a driver (a process or page table) arrives, the attendant assigns them the next available numbered spot (`0x1000` , `0x2000` , etc.).

---

2. Architecture & Data Structures



Key Constants

- `PAGE_SIZE: u64 = 4096` : Standard x86_64 base page size (4 KiB).
- `MAX_REGIONS: usize = 64` : Maximum number of non-contiguous usable RAM blocks tracked.

---

3. Struct Details

`MemoryRegion`

Represents a contiguous block of usable physical memory reported by firmware:

```
#[derive(Debug, Clone, Copy)]
pub struct MemoryRegion {
    pub start: PhysAddr,
    pub page_count: u64,
}
```

`PhysicalMemoryManager`

The state tracker for physical memory:

```
pub struct PhysicalMemoryManager {
    regions: [Option<MemoryRegion>; MAX_REGIONS],
    region_count: usize,
    current_region_idx: usize,
    next_page_in_region: u64,
    total_usable_bytes: u64,
    allocated_frames_count: u64,
}
```

---

4. API Reference & Methods

`PhysicalMemoryManager::add_region(&mut self, start: PhysAddr, page_count: u64)`

Registers a contiguous block of usable physical memory into the tracker.

`PhysicalMemoryManager::allocate_frame(&mut self) -> Option<PhysFrame<Size4KiB>>`

Finds the next free 4 KiB frame from the current memory region and advances the bump pointer. If a region is exhausted, it moves to the next region. - Returns `Some(PhysFrame)` containing the physical start address. - Returns `None` if physical memory is exhausted (Out of Memory).

`PhysicalMemoryManager::total_usable_memory(&self) -> u64`

Returns total detected usable physical RAM in bytes.

`PhysicalMemoryManager::allocated_memory(&self) -> u64`

Returns total allocated physical RAM in bytes.

`PhysicalMemoryManager::free_memory(&self) -> u64`

Returns currently unallocated physical RAM in bytes.

`x86_64::structures::paging::FrameAllocator<Size4KiB>` Trait Implementation

```
unsafe impl FrameAllocator<Size4KiB> for PhysicalMemoryManager {
    fn allocate_frame(&mut self) -> Option<PhysFrame<Size4KiB>> {
        self.allocate_frame()
    }
}
```

Enables `PhysicalMemoryManager` to serve directly as the frame supplier for `x86_64` page table manipulators and memory allocators.

---

5. UEFI Firmware Integration & Low Memory Protection

```
pub fn init_memory_from_uefi()
```

Queries `uefi::boot::memory_map(MemoryType::LOADER_DATA)`: 1. Iterates through all memory descriptors provided by the UEFI firmware. 2. Filters specifically for `uefi::boot::MemoryType::CONVENTIONAL`. 3. **Reserves the First 1 MiB (`0x100000`)**: Skips addresses below 1 MiB to protect legacy Real-Mode Interrupt Vector Tables (IVT), BIOS Data Areas (BDA), and to prevent `PhysAddr(0)` from being issued as a valid frame (protecting against null-pointer aliasing). 4. Registers each usable conventional memory block into the global `FRAME_ALLOCATOR` mutex.

---

6. Practical Example

```
use crate::memory::frame_allocator::{FRAME_ALLOCATOR, init_memory_from_uefi};

// 1. Read UEFI firmware memory map and discover usable RAM
init_memory_from_uefi();

// 2. Allocate a 4 KiB physical frame
let frame = {
    let mut alloc = FRAME_ALLOCATOR.lock();
    alloc.allocate_frame().expect("System has available RAM")
};

// 3. Inspect frame address
let physical_address = frame.start_address().as_u64();
```

STAGE 5: VIRTUAL MEMORY • BOOT SEQUENCE ORDER #11

Virtual Memory & Paging Documentation

- **Source File**: `src/memory/paging.rs`
- **Subsystem**: Memory Management (Virtual Layer & Hardware MMU)

---

1. Conceptual Overview

Modern 64-bit CPUs do not let user programs or even high-level kernel functions read and write directly to physical RAM chips. Instead, they use **Virtual Memory**.

Every address a program interacts with is a **Virtual Address**. The hardware Memory Management Unit (MMU) consults a multi-level tree of translation tables (called **Page Tables**) to convert that virtual address into a **Physical Address** in RAM.

Benefits of Paging

1. **Isolation**: Process A and Process B can both have code at virtual address `0x400000`, but they map to completely different physical RAM chips.
2. **Protection**: Pages can be marked read-only, non-executable (NX/XD bit to prevent malware buffer-overflow attacks), or kernel-only.
3. **Continuity**: Large arrays can appear contiguous in virtual memory even if their physical frames are scattered across RAM.

---

2. 4-Level Paging Architecture (x86_64)

In 64-bit x86 architecture, translating a 48-bit canonical virtual address uses a 4-level tree:

```
Virtual Address:
[ PML4 Index (9 bits) | PDPT Index (9 bits) | PD Index (9 bits) | PT Index (9 bits) | Offset (12 b
        |                      |                    |                  |                    |
        v                      v                    v                  v                    v
   [ CR3 Register ]
        |
        +---> PML4 Table (Level 4)
                  |
                  +---> PDPT Table (Level 3 - Page Directory Pointer Table)
                            |
                            +---> PD Table (Level 2 - Page Directory)
                                      |
                                      +---> PT Table (Level 1 - Page Table)
                                                |
                                                +---> Physical Frame (4 KiB) + Offset
```

- **CR3**: The CPU control register holding the physical address of the active Level 4 (PML4) table.

3. The `AddressSpace` Struct

```
pub struct AddressSpace {
    pml4_phys: PhysAddr,
}
```

Represents a virtual memory context anchored by its PML4 root table.

---

4. API Reference & Methods

`AddressSpace::active() → Self`

Reads the CPU's `CR3` register to attach to the currently running address space.

```
let active_space = AddressSpace::active();
```

`AddressSpace::pml4_physical_address(&self) → PhysAddr`

Returns the physical address of the Level 4 Page Table.

`AddressSpace::table_at<'a>(&self, phys: PhysAddr) → &'a PageTable`

(Unsafe) Dereferences a physical address as an active `PageTable` reference. In native 64-bit UEFI environments, memory is identity-mapped, allowing direct address dereferencing during bootstrap.

`AddressSpace::translate_addr(&self, addr: VirtAddr) → Option<PhysAddr>`

Walks the 4-level page table hierarchy for a given virtual address: 1. Checks PML4 entry for `PRESENT`. 2. Checks PDPT entry for `PRESENT` and handles 1 GiB huge pages. 3. Checks PD entry for `PRESENT` and handles 2 MiB huge pages. 4. Checks PT entry for `PRESENT`. 5. Extracts the physical frame base and adds the 12-bit offset (`0..4095`). - Returns `Some(PhysAddr)` if the address is currently mapped. - Returns `None` if any table level in the walk is unmapped.

---

5. Practical Example

```
use crate::memory::paging::AddressSpace;
use x86_64::VirtAddr;

let space = AddressSpace::active();
let virt_ptr = VirtAddr::new(0x1000_5000);

// Translate to physical address:
match unsafe { space.translate_addr(virt_ptr) } {
    Some(phys) => println!("Virtual {:?} maps to Physical {:?}", virt_ptr, phys),
    None => println!("Virtual {:?} is not mapped in this address space", virt_ptr),
}
```

STAGE 6: SECURITY & CAPABILITIES • BOOT SEQUENCE ORDER #12

Capability Subsystem Overview

- **Source File**: `src/capability/mod.rs`
- **Subsystem**: Capability-Based Security

---

1. Conceptual Overview

Traditional operating systems (like standard Linux or Windows) often rely on **Access Control Lists (ACLs)** or ambient authority (e.g., "this process runs as root/administrator, so it can touch anything"). If an administrative process has a bug or is hijacked, the entire system is compromised.

Hydra uses Capability-Based Security (similar to seL4 and Fuchsia Zircon).

The Real-World Analogy

Think of a **Capability** as a non-forgeable, digital smart keycard: - You cannot just claim "I am the administrator". - You can only perform an operation if you hold a specific keycard that explicitly grants the right to perform that action on that exact resource. - You can hand a copy of a keycard with restricted permissions to another process (e.g., giving someone a key that allows reading a file, but not modifying or deleting it).

---

2. The `Rights` Bitflags

`Rights` defines what actions a process holding a capability is permitted to execute:

```
#[derive(Debug, Clone, Copy, PartialEq, Eq)]
pub struct Rights(pub u32);
```

Supported Rights Flags

Flag	Value	Description
`Rights::READ`	`1 << 0` (1)	Permission to inspect data in the target object.
`Rights::WRITE`	`1 << 1` (2)	Permission to modify or overwrite the target object.
`Rights::EXECUTE`	`1 << 2` (4)	Permission to execute machine instructions from the object (memory frame).
`Rights::GRANT`	`1 << 3` (8)	Permission to delegate, duplicate, or transfer this capability to another process.
`Rights::CALL`	`1 << 4` (16)	Permission to invoke or call an IPC Endpoint.
`Rights::ALL`	`0x1F` (31)	Full combined permissions (`READ | WRITE | EXECUTE | GRANT | CALL`).

Checking Permissions

- `rights.contains(required)` : Returns `true` if all bits in `required` are present.

---

3. The `CapType` Enum

`CapType` defines what kind of physical or virtual kernel resource the capability represents:

```
#[derive(Debug, Clone, Copy, PartialEq, Eq)]
pub enum CapType {
    UntypedMemory { size_bytes: usize },
    MemoryFrame { physical_addr: u64 },
    AddressSpace { root_cr3: u64 },
    Endpoint { endpoint_id: u32 },
    Thread { thread_id: u32 },
    HardwareDevice { device_id: u32 },
}
```

- `UntypedMemory` : Raw, unallocated physical memory blocks that can be carved up into specific objects.
- `MemoryFrame` : A specific 4096-byte page of physical RAM.
- `AddressSpace` : A virtual memory context (represented by the x86_64 CR3 root page table).
- `Endpoint` : A communication mailbox for IPC.
- `Thread` : A schedulable execution thread.
- `HardwareDevice` : An I/O hardware device (such as a GPU, network card, or timer).

---

4. The `Capability` Struct

```rust
#[derive(Debug, Clone, Copy)]
pub struct Capability {
    pub cap_type: CapType,
    pub rights: Rights,
}
```

Combines the resource descriptor (`cap_type`) with the allowable permissions (`rights`).

Constructor

```rust
let memory_cap = Capability::new(
    CapType::MemoryFrame { physical_addr: 0x1000 },
    Rights(Rights::READ.0 | Rights::WRITE.0)
);
```

STAGE 6: SECURITY & CAPABILITIES • BOOT SEQUENCE ORDER #13

Capability Table Module Documentation

- **Source File**: `src/capability/table.rs`
- **Subsystem**: Capability-Based Security

---

1. Conceptual Overview

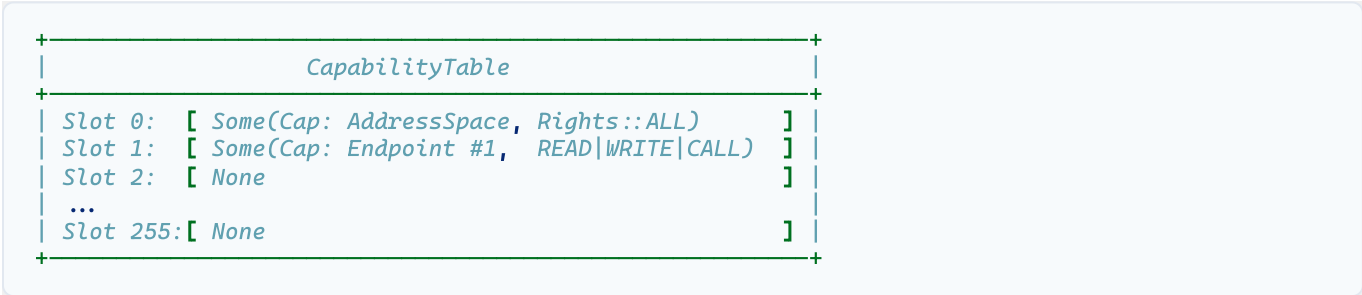
In Hydra, userland processes cannot store raw pointers to kernel objects or hardware devices. Instead, the kernel maintains a **Capability Table** (also known as a **CSpace** or **CNode**) on behalf of each process.

The Real-World Analogy

Think of a **Capability Table** like a keyring or safe-deposit locker: - The process holds numbers (slots: `0`, `1`, `2`, ...). - When a process asks the kernel, "*Read from memory at Slot 3*", the kernel checks Slot 3 in that process's Capability Table. - If Slot 3 contains a valid capability and includes `Rights::READ`, the kernel performs the operation. - If the slot is empty or lacks the required right, the kernel immediately denies the request. - The process never touches the raw physical address or hardware pointer directly, preventing spoofing and unauthorized access.

---

2. Architecture & Constants



- `MAX_CAPABILITIES: usize = 256`: Each table can hold up to 256 active capabilities per task in fixed array storage, avoiding dynamic heap allocation.

---

3. The `CapabilityTable` Struct

```
pub struct CapabilityTable {
    slots: [Option<Capability>; MAX_CAPABILITIES],
    next_free_slot: usize,
}
```

- `slots`: Fixed array of 256 optional capability slots.
- `next_free_slot`: Circular search cursor to quickly find available slots upon insertion.

---

4. API Reference & Methods

`CapabilityTable::new() -> Self`

Creates a new, empty capability table where all 256 slots are `None`.

`CapabilityTable::insert(&mut self, cap: Capability) -> Result<usize, &'static str>`

Allocates the first available empty slot, stores the capability, and returns the slot index (`0..255`). - **Returns** `Ok(slot_index)` on success. - **Returns** `Err("Capability table full")` if all 256 slots are currently occupied.

`CapabilityTable::get(&self, slot: usize) -> Option<&Capability>`

Retrieves a read-only reference to the capability stored at the specified slot. - Returns `None` if the slot is out of bounds or empty.

`CapabilityTable::revoke(&mut self, slot: usize) -> Option<Capability>`

Removes and clears the capability at `slot`, effectively revoking the process's access to that resource.

`CapabilityTable::check_rights(&self, slot: usize, required_rights: Rights) -> bool`

Helper method that verifies: 1. Whether `slot` contains an active capability. 2. Whether the capability possesses all bits defined in `required_rights`.

---

5. Practical Example

```rust
use crate::capability::table::CapabilityTable;
use crate::capability::{Capability, CapType, Rights};

let mut cspace = CapabilityTable::new();

// 1. Grant access to a hardware memory frame
let frame_cap = Capability::new(
    CapType::MemoryFrame { physical_addr: 0x2000_0000 },
    Rights(Rights::READ.0 | Rights::WRITE.0)
);

// 2. Insert into the process table
let slot = cspace.insert(frame_cap).expect("Table has free slots");

// 3. Verify permissions before performing an operation
if cspace.check_rights(slot, Rights::WRITE) {
    // Permission granted: perform write
} else {
    // Permission denied: trigger error
}
```

STAGE 7: IPC SUBSYSTEM • BOOT SEQUENCE ORDER #14

IPC Subsystem Overview

- **Source File**: `src/ipc/mod.rs`
- **Subsystem**: Microkernel IPC (Inter-Process Communication)

---

1. Role in Hydra OS

The `ipc` module binds together the fundamental primitives for Inter-Process Communication in the Hydra microkernel.

Because Hydra isolates all system services (network stack, file systems, device drivers, cloud node daemons) into separate userland processes, fast and secure message passing is the backbone of the entire operating system.

---

2. Module Organization

- `endpoint.rs`: Provides the `Endpoint` and `SharedEndpoint` message queues.
- `message.rs`: Defines the structure of an IPC `Message`, including inline arguments and security capability tokens.

---

3. Re-exported Types

```
pub use endpoint::Endpoint;
pub use message::Message;
```

These re-exports allow any other module in the kernel to conveniently import IPC types directly via `use crate::ipc::{Endpoint, Message};`.

STAGE 7: IPC SUBSYSTEM • BOOT SEQUENCE ORDER #15

IPC Message Module Documentation

- **Source File**: `src/ipc/message.rs`
- **Subsystem**: Microkernel IPC (Inter-Process Communication)

---

1. Conceptual Overview

In a microkernel operating system like **Hydra**, core operating system services (such as file systems, network drivers, and device managers) run as separate, isolated programs rather than inside a single monolithic kernel. Because these isolated programs cannot directly access each other's memory, they must communicate by sending **Messages** across a secured boundary.

The Real-World Analogy

Think of an **IPC Message** like a registered letter or courier package: 1. **The Label (`label`)**: The subject line or action request (e.g., "Read File", "Get Time", "Allocate Memory"). 2. **The Fast Payload (`args`)**: A quick handwritten note with up to 4 numbers or pointers that fit directly into the CPU's registers for high-speed delivery without touching RAM. 3. **The Enclosed Key/Passport (`capability`)**: An optional security token granted to the receiver, giving them special permission to access a specific memory buffer, device, or channel.

---

2. Architecture & Data Structures

```
+-------------------------------------------------------------+
|                         Message                             |
+-------------------------------------------------------------+
| label: u32                | Protocol / Syscall ID           |
| args: [u64; 4]            | Up to 4 fast 64-bit parameters  |
| capability: Option<Cap>   | Optional transferred security   |
|                           | token / permission              |
+-------------------------------------------------------------+
```

Constants

- `MAX_INLINE_WORDS: usize = 4`: The number of 64-bit arguments (`u64`) carried directly within a message. Using 4 words (32 bytes) allows the microkernel to transfer arguments directly using CPU registers (such as `RDI`, `RSI`, `RDX`, `RCX` on x86_64), making IPC fast.

---

3. The `Message` Struct

```rust
#[derive(Debug, Clone, Copy)]
pub struct Message {
    pub label: u32,
    pub args: [u64; MAX_INLINE_WORDS],
    pub capability: Option<Capability>,
}
```

Fields Explained

Field	Type	Description
`label`	`u32`	An operation code indicating what action is requested (e.g. `0x1001` for ping, `0x2001` for read).
`args`	`[u64; 4]`	Array of 4 inline 64-bit numbers. Can be integers, memory offsets, or byte lengths.
`capability`	`Option<Capability>`	An optional security token transferred along with the message to delegate authority.

---

4. API Reference & Methods

`Message::new(label: u32) → Self`

Creates a new message with the given operation label. All 4 arguments are initialized to `0`, and no capability is attached.

```rust
let msg = Message::new(0x01); // Simple ping message
```

`Message::with_args(label: u32, args: [u64; MAX_INLINE_WORDS]) → Self`

Creates a new message with an operation label and up to 4 inline arguments.

```rust
// Example: requesting to read 512 bytes starting at offset 1024
let msg = Message::with_args(0x1002, [1024, 512, 0, 0]);
```

`Message::with_cap(mut self, cap: Capability) → Self`

Builder pattern method to attach a capability to an existing message. This transfers access rights to the receiving process.

```rust
let msg = Message::new(0x2000).with_cap(shared_memory_cap);
```

---

5. Practical Example

```
use crate::capability::{Capability, CapType, Rights};
use crate::ipc::Message;

// 1. Create a capability granting READ access to an endpoint
let ep_cap = Capability::new(CapType::Endpoint { endpoint_id: 10 }, Rights::READ);

// 2. Prepare an IPC message carrying data and delegating the capability
let message = Message::with_args(0x1001, [42, 1337, 0, 0])
    .with_cap(ep_cap);

// 3. The message is now ready to be sent over an Endpoint!
```

STAGE 7: IPC SUBSYSTEM • BOOT SEQUENCE ORDER #16

IPC Endpoint Module Documentation

- **Source File**: `src/ipc/endpoint.rs`
- **Subsystem**: Microkernel IPC (Inter-Process Communication)

---

1. Conceptual Overview

In a microkernel architecture, programs cannot talk directly to each other because their address spaces are completely isolated by the CPU's memory management unit (MMU). An **Endpoint** acts as a meeting place (rendezvous point) or postal mailbox managed by the kernel.

The Real-World Analogy

Imagine an apartment building where residents cannot enter each other's apartments: - An **Endpoint** is a secure PO Box in the lobby. - Process A drops a `Message` into the box (`send`). - Process B checks the box and retrieves the message (`receive`). - The kernel ensures only residents holding the correct security key (`Capability`) can post or retrieve items from that specific box.

---

2. Architecture & Data Structures



Queue Parameters

- `QUEUE_CAPACITY: usize = 16` : The maximum number of unread messages the endpoint buffer can hold before rejecting new incoming messages with an error.

---

3. Struct Details

`Endpoint`

The core internal endpoint managing a circular ring buffer of messages.

```
pub struct Endpoint {
    id: u32,
    buffer: [Option<Message>; QUEUE_CAPACITY],
    head: usize,
    tail: usize,
    count: usize,
}
```

- `id`: A unique numerical identifier for this endpoint.
- `buffer`: Fixed-size array holding pending messages without needing dynamic heap allocations.
- `head`: Read pointer pointing to the next message to be consumed (FIFO order).
- `tail`: Write pointer pointing to the next free slot for an incoming message.
- `count`: Current number of queued, unread messages.

`EndpointState`

Indicates the scheduling and readiness state of the communication channel:

```
#[derive(Debug, Clone, Copy, PartialEq, Eq)]
pub enum EndpointState {
    Idle,
    Active,
}
```

`SharedEndpoint`

A thread-safe wrapper around `Endpoint` using a `spin::Mutex`. This allows concurrent threads and interrupts to safely send and receive messages without race conditions.

```
pub struct SharedEndpoint(Mutex<Endpoint>);
```

---

4. API Reference & Methods

`Endpoint::new(id: u32) -> Self`

Initializes a new empty message queue with the specified ID.

`Endpoint::send(&mut self, msg: Message) -> Result<(), &'static str>`

Pushes a message into the queue: - **Returns** `Ok(())` if the message was placed into the buffer. - **Returns** `Err("Endpoint message queue is full")` if all 16 slots are occupied.

`Endpoint::receive(&mut self) -> Option<Message>`

Pulls the oldest message out of the queue (FIFO): - **Returns** `Some(Message)` if a message is available. - **Returns** `None` if the queue is currently empty.

`Endpoint::pending_count(&self) -> usize`

Returns how many messages are waiting to be read.

`SharedEndpoint::new(id: u32) → Self`

Constructs a new mutex-protected endpoint.

`SharedEndpoint::send(&self, msg: Message) → Result<(), &'static str>`

Acquires the spinlock and sends a message into the underlying endpoint.

`SharedEndpoint::receive(&self) → Option<Message>`

Acquires the spinlock and consumes the next available message.

`SharedEndpoint::id(&self) → u32`

Returns the ID of the wrapped endpoint.

---

5. Practical Example

```
use crate::ipc::{Endpoint, Message};

// 1. Create a communication endpoint with ID 1
let mut mailbox = Endpoint::new(1);

// 2. Sender prepares and sends a message
let request = Message::with_args(0x100, [1, 2, 3, 4]);
mailbox.send(request).expect("Queue has space");

// 3. Check queue status
assert_eq!(mailbox.pending_count(), 1);

// 4. Receiver reads the message
if let Some(msg) = mailbox.receive() {
    // Process msg.label and msg.args ...
}
```

STAGE 8: FILE SERVER & STORAGE • BOOT SEQUENCE ORDER #17

ZFS Microkernel File Server Documentation

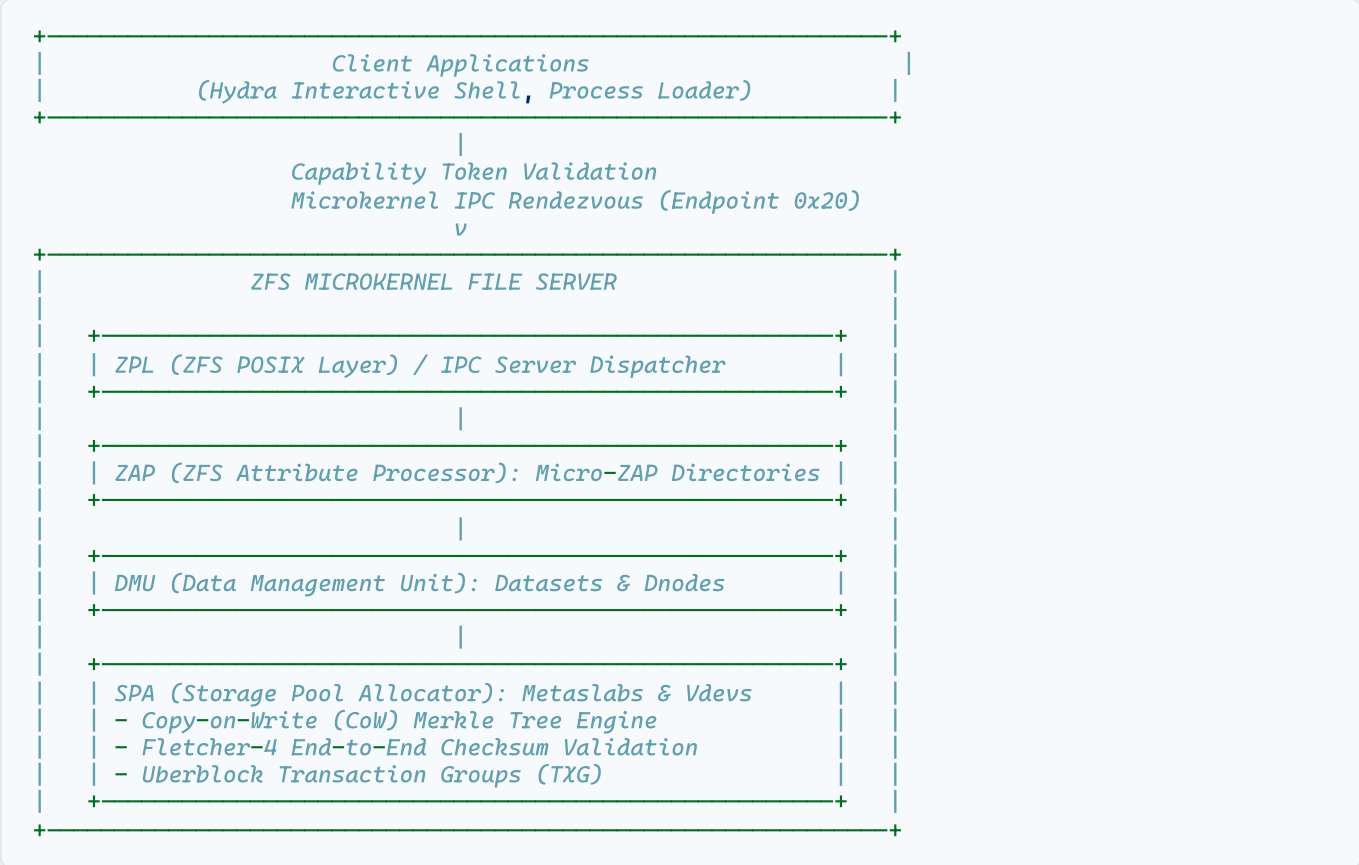
- **Source Directory**: `src/fs/zfs/` & `src/fs/client.rs`
- **Subsystem**: Default Filesystem Architecture & Standalone File Server
- **Endpoint**: `0x20` (`ENDPOINT_ZFS_SERVER`)

---

1. Architectural Motivation

In traditional monolithic operating systems, the filesystem runs in kernel space (Ring 0) with direct block device access. This violates principle-of-least-privilege and microkernel design tenants. A single bug or corruption in the filesystem driver can compromise the entire operating system kernel.

Hydra implements **ZFS (Zettabyte File System)** as the default filesystem, structured strictly as an **autonomous Microkernel File Server (`ZfsServer`)**.



---

2. ZFS Core Architecture

Hydra's ZFS implementation faithfully models the fundamental layers of OpenZFS:

1. Storage Pool Allocator (SPA)

- **Storage Pool**: Named `hydrapool` by default.
- **VDEV (Virtual Device)**: Manages physical block storage or memory-backed frames.
- **Copy-on-Write (CoW)**: Data blocks are never updated in-place. Writing allocates new blocks, preserves previous data for snapshots, and links to a new Merkle root.
- **Active Uberblock**: The root of the storage pool (`magic: 0x00bab10c` , `version: 5000`). Every write transaction commits an incremented Transaction Group (`TXG`).

2. Fletcher-4 End-to-End Checksums

Every block pointer (`blkptr_t`) carries a 256-bit Fletcher-4 checksum (4 parallel 64-bit accumulators): $$\text{Fletcher-4: } a = \sum w_i, \quad b = \sum a_i, \quad c = \sum b_i, \quad d = \sum c_i$$ Upon read operations, the ZFS File Server recalculates the checksum and detects any silent data corruption before returning data to the caller.

3. Data Management Unit (DMU) & Dnodes

- Objects in ZFS are represented by `DnodePhys` structures (`obj_type` , block size, file size, direct block pointers).
- Supported Object Types:
- `PlainFile` : Data payload blocks.
- `Directory` : Associated with a ZAP object.
- `Dataset` : Isolated filesystem namespace (`hydrapool/root` , `hydrapool/bin`).
- `MasterNode` : Meta-Object Set (MOS) root.

4. ZFS Attribute Processor (ZAP)

- Micro-ZAP structure indexing directory entries (mapping string filenames to object IDs).
- Enables $O(1)$ file lookups inside datasets.

---

3. Microkernel IPC Protocol (`Endpoint #0x20`)

Clients interact with the File Server by sending `Message` envelopes through `ENDPOINT_ZFS_SERVER` .

Protocol Label	Opcode	Arguments (`args`)	Return Values	Description
`ZFS_IPC_PING`	`0x3001`	-	`[status, txg, 0, 0]`	Server health check and active TXG.
`ZFS_IPC_LOOKUP`	`0x3002`	`[dir_obj, name_hash, 0, 0]`	`[status, obj_id, obj_type, 0]`	Resolves filename to object ID.
`ZFS_IPC_READ`	`0x3003`	`[obj_id, dest_buf_ptr, max_len, 0]`	`[status, bytes_read, txg, 0]`	Reads and checksum-verifies file data.
`ZFS_IPC_WRITE`	`0x3004`	`[obj_id, src_buf_ptr, len, 0]`	`[status, new_txg, 0, 0]`	CoW writes data and commits new TXG.
`ZFS_IPC_POOL_STATUS`	`0x3006`	-	`[status, total_mb, alloc_kb, free_txg]`	Reports pool metrics and health.
`ZFS_IPC_SNAPSHOT`	`0x3007`	`[dataset_id, 0, 0, 0]`	`[status, txg, 0, 0]`	Creates point-in-time snapshot.

---

4. Capability-Based Security Enforcement

The File Server verifies client capabilities on every request: - **Read Requests**: The client must present a capability containing `Rights::READ`. If missing, the server responds with `ZFS_STATUS_PERMISSION_DENIED`. - **Write / CoW Transactions**: Requires `Rights::WRITE`. - **Snapshots**: Requires `Rights::WRITE`.

---

5. Interactive Shell Commands

Hydra provides native user commands to interact with the ZFS File Server:

```
hydra> zpool status
  pool: hydrapool
 state: ONLINE
status: The pool is operating normally via the Microkernel ZFS File Server.
action: No known data errors detected; all blocks checksum-verified.
config:
        NAME                STATE     READ WRITE CKSUM
        hydrapool           ONLINE       0     0     0
          vdev-ramdisk-0    ONLINE       0     0     0

Pool Properties & Metrics:
  - Capacity:           8 MiB
  - Allocated:          128 KiB
  - Free Space:         8064 KiB
  - Active Uberblock:   TXG #2
  - Checksum Algorithm: Fletcher-4 (4x 64-bit words)
  - Allocation Policy:  Copy-on-Write (CoW) Merkle Tree

hydra> zfs list
NAME                     USED    AVAIL MOUNTPOINT
-------------------------------------------------------
hydrapool/root           128K    8064K /
hydrapool/bin            128K    8064K /bin

hydra> zfs cat readme.txt
Hydra OS Default Filesystem: OpenZFS Architecture
Integrated as an independent Microkernel File Server.
All operations are verified via Fletcher-4 checksums!

hydra> zfs write demo.txt "Hello from ZFS on Hydra Microkernel!"
[+] CoW write committed to ZFS File Server!
    - File: 'demo.txt' (36 bytes)
    - Merkle Root Updated to TXG #3
    - Fletcher-4 Checksum computed & stored
```

STAGE 9: INTERACTIVE SHELL • BOOT SEQUENCE ORDER #18

Interactive Microkernel Shell Documentation

- **Source Directory**: `src/shell/`
- **Subsystem**: Microkernel User Interface & Command REPL

---

1. Conceptual Overview

Operating system development often starts with non-interactive boot routines that print diagnostic logs and halt. To enable active exploration, debugging, and experimentation with kernel primitives on live hardware or virtual machines, Hydra provides an **Interactive Microkernel Shell**.

The shell is architected as a modular, extensible command registry where every command or tool lives in its own dedicated source file:

```
src/shell/
├── mod.rs                  # REPL loop, banner, prompt formatting, and command dispatcher
├── state.rs                # ShellState (CSpace, IPC endpoint, prompt color, cwd, history)
├── command.rs              # The `Command` trait and argument parsing helpers
├── editor.rs               # Interactive line editor (history cycling, tab completion, arrow keys
├── history.rs              # Command history ring buffer (32 entries)
└── commands/               # Modular Command Implementations
    ├── mod.rs              # Static command registry (`COMMANDS`)
    ├── gnu_fs.rs           # GNU Filesystem utils (`pwd`, `cd`, `mkdir`, `rmdir`, `rm`, `cp`, `mv
    ├── gnu_text.rs         # GNU Text inspection (`head`, `tail`, `wc`, `grep`, `basename`, `dirn
    ├── gnu_sys.rs          # GNU System & scripting (`uname`, `whoami`, `sleep`, `seq`, `yes`, `t
    ├── fs_cmds.rs          # File operations (`ls`, `cat`, `write`)
    ├── history_cmd.rs      # Session history viewer (`history`)
    ├── script.rs           # Bash / POSIX script runner (`sh`, `source`, `.`)
    ├── help.rs             # Help command & usage manual generator
    ├── info.rs             # OS version & architecture overview (`info`, `version`)
    ├── status.rs           # Subsystem health inspection
    ├── cpu.rs              # CPU control registers (`cpu`)
    ├── mem.rs              # Memory statistics & allocation (`mem`, `alloc`)
    ├── translate.rs        # Virtual-to-physical paging translation (`translate`)
    ├── caps.rs             # CSpace inspection & revocation (`caps`, `cap_insert`, `cap_revoke`)
    ├── ipc.rs              # Microkernel message rendezvous (`ipc`)
    ├── time.rs             # Real-Time Clock (`time`, `date`)
    ├── power.rs            # ACPI/UEFI power management (`shutdown`, `reboot`, `halt`)
    ├── zfs_cmds.rs         # OpenZFS administration (`zpool`, `zfs`)
    ├── efifs.rs            # Low-level UEFI FAT boot volume diagnostics (`efifs`)
    └── misc.rs             # Utilities (`color`, `echo`, `clear`, `panic`)
```

Zsh-Style Interactive Line Editor & History

The shell includes an interactive line editor (`editor.rs`): - **Up / Down Arrow History Navigation**: Traverses the command history buffer (`history.rs`). Pressing ↑ recalls previously entered commands, while ↓ cycles forward or restores the typed draft line. - **Tab Auto-Completion (Zsh Style)**: Pressing Tab auto-completes built-in command names, aliases, and filesystem paths in the current directory. When multiple matches exist, a color-

coded column listing is displayed, and the prompt and current editing line are cleanly reprinted. - **In-Line Cursor Navigation**: ← and → move the cursor anywhere within the line; `Home` jumps to the start, `End` jumps to the end. - **In-Line Editing**: Character insertion, `Backspace`, and `Delete` work anywhere in the line with automatic redraw. - **History Expansion** `!!`: Typing `!!` echoes and immediately re-executes the most recent command.

Standard Terminal Key Combinations

The interactive line editor and execution runtime implement standard POSIX / Readline / Zsh control key bindings:

Key Combination	Action / Function	Behavior
`Ctrl+C`	**Interrupt / Cancel**	While editing: cancels current input, prints `^C`, and restarts prompt on a new line. While running a program (`yes`, `sleep`, `seq`, `sh`, `cat`, `grep`, `head`, `tail`): halts execution immediately with status `130`.
`Ctrl+D`	**EOF / Forward Delete**	If the line buffer is empty: signals EOF (`logout` / clean shutdown). If the line is non-empty: deletes character under cursor (same as `Delete`).
`Ctrl+L`	**Clear Screen**	Clears the UEFI display and redraws the shell prompt and current line buffer at row 0.
`Ctrl+A`	**Line Start**	Jumps cursor to the beginning of the line (equivalent to `Home`).
`Ctrl+E`	**Line End**	Jumps cursor to the end of the line (equivalent to `End`).
`Ctrl+U`	**Kill Line Before**	Deletes all text from the start of the line up to the cursor.
`Ctrl+K`	**Kill Line After**	Deletes all text from the cursor to the end of the line.
`Ctrl+W`	**Kill Word Backward**	Erases the previous whitespace-delimited word before the cursor.
`Ctrl+P`	**Previous History**	Cycles backward in command history (equivalent to ↑).
`Ctrl+N`	**Next History**	Cycles forward in command history (equivalent to ↓).
`Ctrl+B`	**Move Backward**	Moves cursor left by one character (equivalent to ←).
`Ctrl+F`	**Move Forward**	Moves cursor right by one character (equivalent to →).
`Ctrl+H`	**Backspace**	Deletes character to the left of the cursor.
`Ctrl+Z`	**Suspend**	Displays `^Z` notification.

Shell Script Runner (Bash Compatible)

The shell provides a built-in script interpreter (`sh` , `source` , `.`) capable of executing `.sh` files stored in ZFS or EFI boot volumes: - Supports `#` comment lines and shebangs (`#!/bin/sh` , `#!/bin/bash`). - Supports command chaining with `;` (e.g. `mkdir /tmp/test; cd /tmp/test; touch a.txt; ls`). - Supports script arguments (`$1` , `$2` , `$0`) and last exit status (`$?`). - Pre-loaded with `/bin/demo.sh` demonstrating automated multi-command execution.

Terminal Scrolling via Scroll Lock & Arrows

Hydra implements classic PC terminal scrolling using the **Scroll Lock** mode and arrow keys:

1. **When Scroll Lock is Active (`[SCROLL]` Prompt)**: - The shell freezes command execution and converts navigation keys into terminal viewport navigation:
    - ↑ (Up Arrow): Scrolls the terminal screen **UP** line-by-line into the 512-line kernel scrollback buffer.
    - ↓ (Down Arrow): Scrolls the terminal screen **DOWN** line-by-line toward the active command prompt.
    - Page Up: Scrolls up by an entire screen page (`rows - 2` lines).
    - Page Down: Scrolls down by an entire screen page.
    - **Full Color & Attribute Preservation**: The scrollback buffer tracks both character bytes and formatting attributes (`(fg & 0xF) | ((bg & 0x7) << 4)`). All syntax highlighting, prompt colors (`LightCyan`), section headers (`Yellow`), directory names (`Green`), and error messages (`LightRed`) are preserved verbatim when scrolling up and down.
    - When scrolled up into history, a cyan status banner displays the visible line range: `-- SCROLL LOCK (up N lines, lines X-Y of Z) [UP/DOWN or PgUp/PgDn, ESC to exit] --` .
    - Allows naturally reviewing long command outputs (like `help` , `cat` , `ls -l` , `history` , or kernel boot diagnostics) that scrolled off the top of the screen without losing any visual hierarchy.
2. **When Scroll Lock is Inactive (Normal Mode)**: - ↑ (Up Arrow) cycles backward through the command history ring buffer. - ↓ (Down Arrow) cycles forward through the command history ring buffer. - Typing any printable character, pressing Enter, or hitting Backspace automatically snaps the viewport back to the active prompt.

3. **Activating & Toggling Scroll Lock**: - **Keyboard Shortcut**: Press Ctrl+S or F12 in the shell to immediately toggle Scroll Lock on/off. The prompt dynamically reflects the mode (`hydra:/bin [SCROLL]>` vs `hydra:/bin>`). - **Direct Page Up**: Pressing Page Up at the prompt automatically enables scrolling mode and pages up through previous screen contents. - **Shell Command**: Run `scroll` (or `scroll on` / `scroll off`) from the command line. - **Exiting Scroll Mode**: Press Escape, Ctrl+S, or Ctrl+C to return to normal prompt mode. - **VirtualBox Note**: In VirtualBox with UEFI enabled, the firmware's `EFI_SIMPLE_TEXT_INPUT_PROTOCOL` does not generate key events for the physical Scroll Lock key (and host OS keyboard hooks often intercept or discard it). Use Ctrl+S, F12, or `scroll on` inside the VM for instant, reliable scrolling.

Dynamic Working Directory Prompt

The shell tracks the current working directory dynamically in `ShellState` (initialized to `/bin`), displaying a standard Unix prompt:

```
hydra:/bin> pwd
/bin
hydra:/bin> cd /
hydra:/> ls
NAME                      TYPE         CHECKSUM     STATUS
──────────────────────────────────────────────────────────
bin                       <DIR>        Fletcher4     ONLINE
root                      <DIR>        Fletcher4     ONLINE
tmp                       <DIR>        Fletcher4     ONLINE
──────────────────────────────────────────────────────────
hydra:/> cd /bin
hydra:/bin>
```

---

2. Command Reference

GNU Core Utilities (POSIX / Linux Compatible)

Command	Syntax	Description	Supported Options & Flags
`pwd`	`pwd`	Prints current working directory.	
`cd`, `chdir`	`cd [dir]`	Changes current working directory. Default: `/root`.	Relative paths, `..`, `.`, `/`
`ls`, `dir`	`ls [path]`	Lists files and directories in ZFS. Default: `cwd`.	`-l` (long format), `-a` (show all)
`cat`, `type`	`cat <file>`	Displays file contents verified with Fletcher-4.	Path resolution relative to `cwd`
`more`, `less`	`more <file>`	Paginates file with interactive Up/Down and PageUp/PageDown scrolling.	`less`, `page` aliases
`write`	`write <file> [text]`	Writes/appends data to a file via CoW transaction.	Path resolution relative to `cwd`
`touch`	`touch <file ... >`	Creates empty file or updates birth TXG / timestamp.	Path resolution relative to `cwd`
`mkdir`	`mkdir <dir ... >`	Creates one or more directories in ZFS.	`-p` (create parents without error)
`rmdir`	`rmdir <dir ... >`	Removes empty directories from ZFS.	Verifies directory empty count
`rm`, `unlink`	`rm <file ... >`	Removes files or directories from ZFS.	`-r` / `-R` (recursive), `-f` (force)
`cp`, `copy`	`cp <src> <dst>`	Copies files within ZFS pool via CoW.	Directory destination resolution
`mv`, `rename`	`mv <src> <dst>`	Moves or renames files or directories in ZFS.	Cross-directory and in-place moves
`stat`	`stat <file>`	Displays detailed file or directory metadata.	Size, blocks, birth TXG, checksum
`df`	`df`	Reports filesystem and storage pool capacity.	`-h` (human-readable: MiB/KiB)

Command	Syntax	Description	Supported Options & Flags
`du`	`du [path]`	Estimates file and directory disk space usage.	`-h` (human-readable), `-s` (summary), `-a`
`head`	`head <file ... >`	Displays first N lines of files. Default: 10 lines.	`-n <num>` or `-<num>`
`tail`	`tail <file ... >`	Displays last N lines of files. Default: 10 lines.	`-n <num>` or `-<num>`
`wc`	`wc <file ... >`	Counts newlines, words, and bytes in files.	`-l` (lines), `-w` (words), `-c` (bytes)
`grep`	`grep <pattern> <file>`	Searches for patterns in files.	`-i` (ignore case), `-n` (line numbers), `-v` (invert), `-c` (count)
`basename`	`basename <path> [suf]`	Strips directory and suffix from filenames.	Optional suffix removal
`dirname`	`dirname <path>`	Strips last component from filename path.	
`uname`	`uname`	Prints operating system and architecture info.	`-a` (all), `-s` (kernel), `-n` (node), `-r` (release), `-v` (version), `-m` (arch)
`whoami`, `id`	`whoami`	Prints current user identity and capabilities.	`-u`, `-g`, `-a` (full capability context)
`sleep`	`sleep <secs>`	Delays execution for specified duration (up to 60s).	Non-blocking firmware stall
`seq`	`seq [FIRST [INC]] LAST`	Prints sequences of integers.	Customizable start, step, and end
`history`, `hist`	`history`	Displays numbered session command history list.	Up/down arrow key cycling
`sh`, `source`, `.`	`sh <script.sh>`	Executes Bash-compatible script files line by line.	`#` comments, `;` chaining, `$1..$9`, `$?`
`!!`	`!!`	Repeats and re-executes the last command.	Shell history expansion
`true`	`true`	Returns success status (exit status 0).	
`false`	`false`	Returns failure status (exit status 1).	

Microkernel & Hardware Diagnostics

Command	Syntax	Description
`help`, `?`	`help [cmd]`	Displays the interactive command manual and specific usage.
`info`, `version`	`info`	Prints Hydra Microkernel OS version, build target, and architecture overview.
`status`	`status`	Reports operational health across GDT, CPU, Frame Allocator, Paging, CSpace, and IPC.
`cpu`	`cpu`	Displays CPU control registers (`CR0`, `CR2`, `CR3` PML4 root, `CR4`).
`mem`, `memory`	`mem`	Displays physical memory manager statistics (Total Usable RAM, Allocated, Free).
`alloc`	`alloc`	Allocates a 4 KiB physical memory frame from the bump allocator and returns its physical address.
`translate`	`translate <hex_virt_addr>`	Walks the active 4-level PML4 page table hierarchy to resolve virtual to physical addresses.
`caps`, `cspace`	`caps`	Inspects the active task's capability table (slot numbers, resource types, and rights bitflags).
`cap_insert`	`cap_insert <ep_id>`	Inserts a new Endpoint capability into the task's CSpace.
`cap_revoke`	`cap_revoke <slot>`	Revokes an existing capability from the specified CSpace slot.
`ipc`	`ipc <label> [a0] [a1]`	Sends and receives a fast register IPC message through rendezvous Endpoint #1.
`time`, `date`	`time`	Queries the hardware Real-Time Clock (RTC) via UEFI runtime services.
`zpool`	`zpool <status|list>`	Inspects and manages ZFS storage pools (`hydrapool`, vdevs, CoW Merkle tree).
`zfs`	`zfs <list|snap|cat|write>`	Manages ZFS datasets (`hydrapool/root`, `hydrapool/bin`, `hydrapool/tmp`) and CoW snapshots.
`efifs`	`efifs <ls|cat|run>`	Low-level firmware diagnostic tool inspecting raw UEFI FAT bootloader media.
`color`	`color <name>`	Customizes shell prompt and text color (`cyan`, `green`, `yellow`, `red`, `white`, `magenta`, `blue`).

Command	Syntax	Description
`scroll`	`scroll [on|off]`	Toggles or sets Scroll Lock mode for Up/Down arrow terminal scrolling.
`echo`	`echo <text ... >`	Echoes input arguments back to the screen.
`clear`, `cls`	`clear`	Clears the active screen buffer and resets the cursor.
`panic`	`panic`	Manually triggers the kernel panic handler to verify diagnostic output.
`reboot`	`reboot`	Reboots the computer via 8042 keyboard pulse, PCI reset, and UEFI cold reset.
`shutdown`	`shutdown`	Powers down the computer via VirtualBox/QEMU ACPI I/O and UEFI shutdown.
`halt`	`halt`	Disables CPU interrupts and enters a low-power `hlt` loop.

---

3. Integrating New Commands & Standalone Programs

Hydra supports two complementary paradigms for adding commands and applications:

Approach A: Dynamic Filesystem Search ($PATH Discovery — No Code Changes!)

This is the recommended approach for integrating standalone applications, external utilities, and third-party binaries.

1. **How It Works**: When a command is entered (e.g. `hydra> matrix` or `hydra> test_app arg1 arg2`), the shell first checks built-in commands. If not found, it dynamically searches the `$PATH` directories on the boot volume: - `\bin\` - `\apps\` - `\EFI\BOOT\` - `\` (root directory)

For each directory, it probes for: - `<dir>\<cmd>.efi` - `<dir>\<cmd>`

2. **Loading & Execution**: - The file is located on the UEFI boot volume via `SimpleFileSystem`. - The binary's PE32+ header (`MZ` signature) is verified. - Memory pages are dynamically allocated and the executable is loaded via `uefi::boot::load_image`. - Arguments are passed through the UEFI `LoadedImage` protocol. - The process is executed via `uefi::boot::start_image`. - Upon termination, control returns immediately to the Hydra interactive shell prompt!

3. **Zero Kernel Modifications**: You can drop any compiled UEFI PE32+ `.efi` executable into `\bin\` or `\apps\` on your boot USB or disk image at any time. **You do NOT need to modify `commands/mod.rs` or recompile the kernel!**

Approach B: Static Built-in Command Integration

Used for core microkernel primitives that require direct access to kernel internals, `CapabilityTable`, or `ShellState`.

Step 1: Create a New Command File

Create a new file in `src/shell/commands/`, e.g. `src/shell/commands/my_tool.rs`:

```
use crate::console::set_color;
use uefi::proto::console::text::Color;
use crate::shell::command::Command;
use crate::shell::state::ShellState;

pub struct MyToolCommand;

impl Command for MyToolCommand {
    fn name(&self) -> &'static str {
        "mytool"
    }

    fn aliases(&self) -> &'static [&'static str] {
        &["mt"]
    }

    fn description(&self) -> &'static str {
        "Demonstrates custom command integration"
    }

    fn usage(&self) -> &'static str {
        "mytool <argument>"
    }

    fn execute(&self, args: &[&str], state: &mut ShellState) {
        set_color(Color::LightGreen, Color::Black);
        println!("[+] MyTool executed with args: {:?}", args);
    }
}
```

Step 2: Register in `src/shell/commands/mod.rs`

In `src/shell/commands/mod.rs`: 1. Add the module: `pub mod my_tool;` 2. Export the struct: `pub use my_tool::MyToolCommand;` 3. Add `&MyToolCommand` to the `COMMANDS` slice: `rust pub static COMMANDS: &[&dyn Command] = &[ ... &MyToolCommand, ];`

---

4. Architecture & Key Features

Thread-Safe Interactive Input

Interactive input is handled by `console::read_line`: - Characters are received from UEFI stdin with a 10 ms sleep interval via `uefi::boot::stall` to ensure low VM CPU utilization. - Real-time backspace support erases characters from screen and buffer seamlessly via cursor positioning. - Enter (`\r` or `\n`) submits the command line.

Multi-Tier Power Management (`shutdown` & `reboot`)

Virtual machine firmware (such as VirtualBox's EFI and QEMU/EDK2) frequently stubs or hangs on generic UEFI runtime `ResetSystem(EfiResetShutdown)` prior to ACPI table activation. Hydra solves this by implementing direct hardware I/O power transitions: 1. **VirtualBox ACPI**: Writes sleep state S5 bits to PIIX3/4 ACPI `PM1a_CNT` (I/O Port `0x4004`). 2. **QEMU ACPI**: Writes to standard PIIX4 ACPI `PM1a_CNT` (I/O Port `0x604`) and debug exit (`0x501`). 3. **Cloud / Bochs ACPI**: Writes to I/O Port `0xb004` and APM (`0xb2`). 4. **Reboot Sequencing**: Pulses the 8042 keyboard controller reset line (`0x64` , `0xFE`) and the PCI reset control register (`0xCF9` , `0x06`) before falling back to UEFI cold reset.

Virtual Address Translation Inspection

The `translate` command allows live debugging of paging:

```
hydra> translate 0x400000
[*] Translating Virtual Address 0x400000 via PML4 tree: Mapped → Physical Address 0x400000
```

On-Demand Memory Allocation

The `alloc` command demonstrates the physical memory frame allocator in real-time:

```
hydra> alloc
[+] Successfully allocated 4 KiB frame at physical address: 0x101000
hydra> mem
Physical Memory Manager Statistics:
  Total Usable RAM:    ...
  Allocated Frames:    ...
```

Live Microkernel IPC

The `ipc` command sends a message through a rendezvous endpoint and reads the delivered payload:

```
hydra> ipc 0x2001 100 200 300
[+] Message queued to Endpoint #1: label=0x2001, args=[100, 200, 300, 0]
[+] Rendezvous complete! Received message from Endpoint #1:
    - Label: 0x2001
    - Args:  [100, 200, 300, 0]
```